Lesson Aim

Develop a procedure to establish a vineyard.

Establishing a New Vineyard

Care and consideration should be taken in selecting a site and establishing a new vineyard. Good planning and site establishment will make management easier and improve chances of vineyard success. Vineyard establishment generally involves several key aspects. These include planning, site preparation, soil conditioning, installing cropping infrastructure such as irrigation and trellises, installation of other infrastructure such as roadways and packing sheds, and plant establishment.

General Planning

Every vineyard needs a plan specifically tailored to the site, the locality, the scale of the enterprise and the owners' requirements. Given that each enterprise has different priorities, and operates under a different set of circumstances, it is impossible to follow a "generic" or standard approach to planning.

You do need to plan all aspects which are important to your operation, including production, finance, land care, facility and equipment management, marketing, product processing (if relevant), and physical layout (i.e. design of the property).

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Planning in each of these areas will involve:

- Determining decisions which need to be made in order to prepare for the future
- Gathering information upon which to base those decisions
- Analysing that information
- Considering the options, and the likely outcome of each option
- Selecting the most appropriate option, based upon the information available
- Acting on the selected option
- Reviewing the situation periodically, and accordingly modifying the action being taken

A well-organised and experienced grower may not need to be too rigid in adhering to a procedure like this; however, many growers will find a real benefit in working through a systematic procedure "on paper".

The surest way to succeed is to move through this procedure step-by-step; writing down everything as you go. This gives you a chance to ponder over what you have written, add new thoughts, or make alterations.

Making Decisions

The grower needs to make decisions every day, but each decision varies in terms of:

Significance

Some decisions will have a greater impact (e.g. buying new land or changing agents). Obviously more significant decisions warrant more careful planning.

Timing

Some decisions need to be made repeatedly and frequently (e.g. managing weeds, irrigating, buying fertiliser or machinery).

Good planning may involve establishing a procedure to help with such decisions (e.g. re-order when the quantity in stock drops to a certain level).

Some decisions need to be made straight away (e.g. a regular weeding schedule, soil moisture monitoring, treatment of a serious disease), while others might not matter if they are delayed (e.g. replacing a fence).

Permanence

Some decisions are more permanent (e.g. planting a new variety), while others are easier to change (e.g. the amount of water applied a particular point in time).

Site Planning

Good planning will assist the viticulturist in developing a vineyard that maximises the effect of positive elements of the site, such as aspect or soil type, and minimises the effect of negative elements, such as frost potential.

Drawing a Vineyard Plan

The first step in a property development strategy is to draw a vineyard plan. Drawn to scale and illustrating important attributes such as contours, creeks or streams, hills, cliffs, and other features, this map will aid in the appropriate exploitation of usable land. Details such as existing trees, river banks, fences, soil types, direction of summer and winter winds, will aid all future planning processes. If you have recently purchased the property, some of this information may be available from previous farm owners, local council departments and even state or national government land departments. Some information will be found by observing conditions on the site.

The inclusion in the map of existing buildings and knowledge of work patterns and access routes will allow efficient time management and will also enable optimum siting of future buildings.

If sheds and the farm residence are placed close together it will reduce time to get from work to home, aid in better husbandry of vines, improve security around the property, and reduce transportation between various components of the property. If it is a new farm location, careful placement of the buildings may even reduce costs. For instance, placing a house close to the road will reduce the connection costs of electricity, water and other services.

The siting of the house and farm buildings should take into account the agricultural value of land, the central location of the house to all useable land, flood levels, etc. Good land should not be built on – it should be reserved for agricultural use, but if the property is flood-prone, then a compromise needs to be made. Check how often the land has been flooded.

The location of farm buildings may also be determined by their use and function in the running of the farm, e.g. manure storage area should be downwind to residences. Consider the downwind land uses in neighbouring parcels of land.

Always remember to allow appropriate thoroughfare for machinery. Ensure safety when driving machinery in terms of overhead powerlines, steep slopes and edges of access tracks (weak shoulders and gullies).

Having a natural source of water on the land is every farmer’s dream. Considering it is actually a resource, many districts require registration for tapping that resource. Extra care is needed when looking at the potential of farm pollution into the waterways by others upstream of your property. Water problems can relate to underground water movement as well as more obvious surface water. Both types of water can bring pollutants from outside of the property.

Note that a licence is usually required to sink a bore or to install a pump from a surface water supply (eg. river or lake), with the volume controlled by the licence requirements.

Water quality is important for crops, animals and humans, and a water purity and quality assessment is therefore recommended. In remote districts, it is crucial to employ water conservation techniques such as water harvesting, selection of dry land species, etc.

Fencing should be placed to optimise land use, assist ease of access through the property, mark off damaged areas (e.g. unstabl ground, erosion-prone soil) and to restrict animal access to vines.

Efficient farming depends a great deal on getting your priorities right, identifying those tasks on the farm which need to be done before other jobs are undertaken. For example, this may mean finishing the vineyard fencing before the shed extension is started, so that animals will not stray and damage the vines.

Money is one of the biggest burdens for the commercial viticulturist and anybody considering a new operation or expanding an existing vineyard must identify their present financial limitation so they can plan the enterprise fully aware of financial constraints. These constraints may slow down the full operation, but it is a safe management strategy to follow. Take one step at a time; do no jump in head first.

Setting Up

Vineyard Layout

The first step in setting up a vineyard is planning how it will be set out. One way to create an effective plan is to first map all of the existing features such as vegetation, buildings, power and water supplies, underground cables, pipes etc. Then work out how the vineyard can best be arranged within the land available. Do not be afraid to consider moving fence lines, or even outbuildings and roads. In some situations it may be appropriate to remove vegetation, but ideally vegetation should be established in other areas to compensate for the loss. This can benefit the property’s operations and can also provide a marketing element.

Square or rectangular plots are easiest to manage and most efficient to use. Tractor operation is easier, access by workers is easier, and there is less wasted space.

Rows can be arranged in various ways:

- On a north-south orientation, to ensure both sides of the vine get equal amounts of sunlight
- Perpendicular to prevailing winds in order to minimise wind damage

- Running down a hill-slope to allow cold air to drain away

In practice, decisions about vine row orientation compromise these considerations with other limitations imposed by the site such as property boundaries or buried cables.

Site Preparation

Site preparation may involve activities such as clearing land and installing roads, water systems and electricity. Care should be taken to conform to local regulations.

While vineyards are frequently planted on hillsides, management of the established vineyard can be made easier by ensuring that the level of the ground is consistent within the row areas of the vineyards. To this end, it may be necessary to use earthmoving equipment to level out the ground during vineyard establishment.

Soil Preparation

Soil preparation requirements can vary significantly, depending on the condition of the soil. There will frequently be a need to level ground and eliminate roots and stones from within the planting area. The grower may also need to control or eradicate weeds prior to planting. Effective weed control in the preparation stage can significantly reduce ongoing labour requirements once the vineyard is established. Effective weed control includes not only eradicating the existing weeds, but also attempting to remove any propagules such as seed or tubers, to minimise the risk of re-infestation.

Eradication can be achieved by various mechanical means, by flame or steam weeding, or with herbicides.

Some soils will require cultivation such as deep ripping to break impermeable layers in the subsoil. Deep ripping will improve drainage make it easier for the roots of the growing vines to penetrate the soil. This is best done when the soil is relatively dry towards the end of summer. (If deep ripping is done in wet soil it can do more harm than good.) Deep ploughing (not rotary hoeing) to 40cm or more should be carried out at this stage. Soil ameliorants, such as lime, compost, or fertilisers, can also be added during this phase of preparation.

Shelter Belts

Shelter belts may need to be planted on large properties, particularly in wind-prone areas. Tall deciduous trees such as Lombardy poplars are well suited for this purpose. Protection is provided for a distance of up to ten times the height of the windbreak, downwind of the shelter belt.

Generally it is better to plant windbreaks during the site establishment phase, as this allows them to mature more quickly. However, newly-planted windbreak plants may be delicate and prone to damage, and may need extra care during their establishment (watering, protection from frost etc); if this is likely to hinder the establishment of the overall site, it may be better to delay planting the windbreak.

Planting the Vines

Planting should be done over winter when the plants are dormant (i.e. without leaves). The vines will most likely be purchased or delivered bare-rooted, which means the roots are bare of soil but wrapped in a protective layer of moist straw or similar material. It is preferable to plant the vines immediately but if this is not possible, they can be kept in a cool area for several days; however the roots must not dry out or be exposed to direct sunlight at any stage during storage. Small quantities of dormant vines can be placed temporarily in a shallow trench or pit, with moist sand used to cover and protect the roots.

At planting time, the vines can be planted into a wide trench (0.9m wide, or 3 feet), as the large zone of loose soil allows the vines to quickly establish roots in their first year of growth. Alternatively the vines are planted into individual holes. Damaged or broken roots are cut off cleanly with sharp secateurs before the plant is placed in the hole or trench. Very long roots can be trimmed back to fit the planting hole but excessive root pruning should be avoided as this removes the plant's energy store.

The backfilled soil should completely cover the roots, but not be placed around the stem as this will lead to basal rots. On grafted plants, the graft union must be 5cm (2inches) above the soil. The exception is in frost-prone areas, where soil is sometimes mounded up around the vine and the graft union when it is planted, to provide protection from frost damage on the emerging buds. Once the buds start growing, the soil is pulled back away from the plant.

The newly-planted vine is watered in thoroughly to moisten and settle the soil, and to remove air pockets around the roots. A layer of organic mulch is beneficial except in frost-prone areas because organic mulches lower soil temperatures and increase the likelihood of frost damage.

The newly-planted vine is trimmed back to three to four buds. No fertilisers are added to the planting hole or placed around the vine, as the vine is dormant and cannot absorb the nutrients.

Vine Spacing

Typical vine spacing is 1.8m (6 feet) apart in rows 3m (9 feet) apart. This allows easy access of machinery between the rows and optimum canopy exposure for the vines. Vines in dry soils and highly vigorous varieties may need wider spacing. European vines are often more closely planted to maximise the yield per acre, although the yield per vine is reduced due to competition.

Cropping Infrastructure

Cropping infrastructure includes structures such as trellises, irrigation systems, sheds to package or process crops, machinery sheds, and other buildings.

It may be possible to establish trellises within the first year of vine growth in some situations. In other situations it may be more practical to establish the trellises prior to planting the vines.

Irrigation systems should ideally be installed prior to planting, as this avoids the possibility of damaging delicate new plants as trenches are dug and pipes are placed in position. In some circumstances, irrigation systems may also be buried. Irrigation mains are frequently installed during the site preparation phase. It will usually be necessary to install pumps and other associated hardware to operate the irrigation system. This may include machinery to supply nutrients with the water if fertigation practices are to be used. Irrigation systems can vary from simple manual systems to complex computerised systems.

The necessity for sheds and other structures will vary according to the size and nature of the vineyard. In some cases it may be possible to delay the construction of some buildings until later in the vineyard development process. For example, it may not be necessary to have processing sheds until the vines are at a stage where they are producing crops.

Equipment

Farm equipment can be a major expense when setting up. If money is a problem, you may consider either hiring or sharing at least some of your equipment. It may not be financially viable for a small operation to own some items of equipment and in many districts, that machinery need may be satisfied by one of the following:

- Hiring a contractor to do the work
- Hiring the machine
- Sharing the machine with a cooperative group
- Buying the machine and hiring it out to recoup some of the costs

Farm equipment pools involve a group of farmers in a local area sharing certain items of machinery, as a cost-saving enterprise. There is little point in three neighbouring properties each purchasing an expensive machine which they use of only four weeks annually. Establishing a vineyard near other vineyards can be beneficial for this reason.

Some of the advantages of sharing include:

- The money saved by each grower can be better used in farm improvements or to purchase other needed equipment.
- Increased value for money, as the item is not superseded before it becomes worn out and needs replacing.
- It costs less to upgrade, because costs are shared.
- Space is saved because there is less machinery on each property (this is particularly significant if the equipment needs to be stored under cover).
- Keeping the machine in a shed is more viable because costs are shared; hence life expectancy of the machine can be extended by undercover storage.
- More frequent use and shared maintenance effort will keep the machine running better (e.g. a machine needs to be used for moving parts to remain free and not seize up).
- Costs might be further off set by hiring the equipment to other growers outside of the equipment pool.

The disadvantages might be that:

- More than one grower might want to use the machine at the same time.
- An amiable agreement on liability, care and use of the equipment might sometimes be difficult to reach.